

ASSADUZZAMAN MUNNA

AI/ML & Software Engineer — Aspiring AI/ML Specialist

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EDUCATION

North South University (NSU) — Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering (Department of ECE)

2022 – May 2026

CGPA: 3.74 / 4.00

EXPERIENCE

AI Engineer

The Data Island — April 2026 – Present

- Promoted from Junior AI Intern to engineer and optimize scalable data pipelines, seamlessly integrating enterprise machine learning frameworks for production systems.

Undergraduate Teaching Assistant (UGA)

North South University (ECE Dept.) — June 2025 – Present

- Mentor undergraduate students by facilitating technical sessions, supporting curriculum delivery, and evaluating engineering assignments.

On-the-Job Training (AI Engineering)

Nippon AI Dojo 2025 (Chowa Giken & AI Samurai Japan) — Sep 2025 – Jan 2026

- Developed practical AI implementation and model optimization skills through hands-on, real-world projects mentored directly by Japanese industry experts.

SKILLS

- **AI & Machine Learning:** PyTorch, TensorFlow, Hugging Face, SAM, MLOps, Knowledge Distillation. NLP, Supervised & Unsupervised Learning, Transformers, Fine-tuning.
- **DevOps & Data:** NVIDIA DeepStream, Docker, Linux (Ubuntu), FFmpeg, SQL, Pandas, NumPy.
- **Software Engineering:** React.js, Next.js, Flutter (Dart), RESTful APIs, System Design, Git.
- **Languages:** Python, C++, JavaScript, Dart, Java, LaTeX.

FEATURED PROJECTS

MiST-ER: Micro-emotion Selective Temporal Emotion Recognition — Ongoing — Deep Learning Architect

- Architecting a lightweight multimodal (Audio/Video/Text) pipeline for micro-emotion classification (achieving 54% accuracy on MESOC) optimized for edge deployment via cross-modal attention mechanisms.

Industrial Automation System (Foil Stamping) — System Architect

- Built a low-latency **Python/OpenCV** backend capable of processing high-throughput inspection streams for industrial manufacturing lines.

PUBLICATIONS

Bone Fracture Detection Using Vision Transformers: A Comparative Analysis of the PiT and CaFormer Models

Atikul Islam Munna, Md. Ibrahim Khalil, **Assaduzzaman Munna**, et al. — 2026 IEEE 2nd Int. Conf. on Secure IoT (SATC), 2026. DOI: 10.1109/SATC69565.2026.11542322

- Evaluated Vision Transformers against CNNs on 4,083 X-ray images, with the PiT model achieving 97.51% testing accuracy in automated fracture diagnosis.

Hybrid CNN-MobileViT Model with Knowledge Distillation for Efficient Renal Calculi Detection

Assaduzzaman Munna, Mushfika Hossain, Anonto Bormon, Riasat Khan. — Submitted to SPICSCON 2026. (Under Review)

- Engineered a knowledge distillation model reducing parameters by **18.5x** for edge deployment, achieving **99.80% test accuracy** with highly optimized inference speeds on an NVIDIA T4 GPU.

ACHIEVEMENTS & CERTIFICATIONS

Nippon AI Dojo 2025 (AI Training Program) — Selected Participant

Completed a rigorous program designed to develop practical AI engineering skills via structured lectures and hands-on projects by Chowa Giken and AI Samurai Japan Bangladesh.